AI HEALTH



Dr Naresh Trehan
CMD, Medanta-The Medicity

MARCH OF **PRECISION** MEDICINE

THROUGH ITS ABILITY TO MAKE SENSE OF HEAPS OF DATA, AI CAN TRANSFORM INDIA'S HEALTHCARE LANDSCAPE BY REDUCING COSTS, IMPROVING EFFICIENCY AND FACILITATING PERSONALISED INTERVENTIONS

The medical community has rapidly acknowledged the potential advantages of Artificial Intelligence (AI) since its emergence in the 1950s. In 1959, it was determined that machines could interpret symptoms diagnostically. By 1960, physicians projected that computers could significantly augment their cognitive abilities by the year 2000. Availability of data through connected devices, increased storage capacities, enhanced computing are a few enablers of AI adoption across industries, including healthcare. From everyday health apps to high-precision analysis of complex medical scans for early diagnosis, to helping tailor treatment plans, AI is useful in drawing insights from a large amount of clinical data for better disease management, prediction and designing outbreak strategies. Machine learning, which involves fitting models to data and learning through training, is utilised extensively in healthcare, especially in precision medicine.

AI is augmenting our ability to think and take considered action as people and clinicians. This holds the promise of significantly contributing towards transforming India's healthcare landscape, which remains riddled with historical challenges, even though healthcare has been one of the biggest success stories of our time. The unstoppable forces of changing demographics, such as an ageing population, altered lifestyle and disease patterns, and climate change, are posing challenges that can't be met through just public-private spending, structural changes in the healthcare network or increasing the number of medical practitioners and allied professionals. Expert use of AI offers us an unparalleled opportunity to surmount challenges both old and new despite increased noise and fear about the machine replacing man across functions.

Health in Your Hands

One of the biggest under-recognised benefits of AI and the Internet of Things (IoT) has been the mass empowerment of people to maintain their health, to the extent possible, on their own. Apps and wearable technology are driving awareness and encouraging healthier lifestyles. The data also helps healthcare professionals trace their patients' behaviour patterns and get a better sense of their health.

AI models are also assisting in early detection and prompt diagnosis of individuals who are at heightened risk of complex chronic illnesses. This helps doctors initiate early treatment, reducing the need for extensive surgical care, and providing specialised interventions. By furnishing critical and pertinent information at the point of care, AI alleviates the cognitive burden on doctors during patient treatment.

For instance, within cardiovascular healthcare, a machine learning model is under development to



Illustration by **NILANJAN DAS** / AI